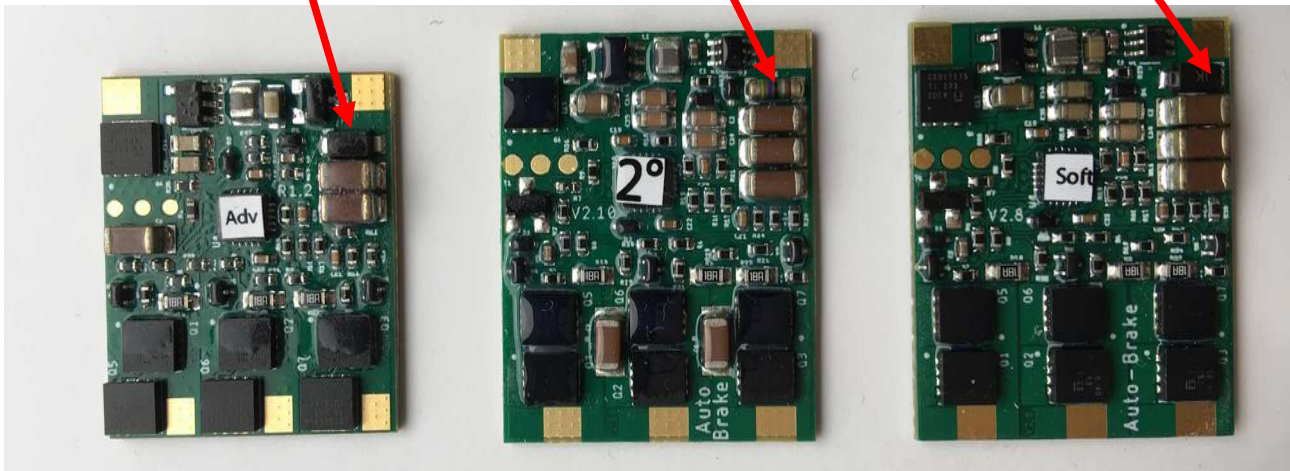


Latvian boards are single side mounted components

code IK

4R7 resistor

code IK



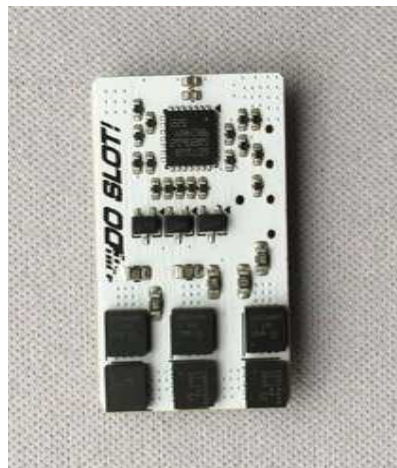
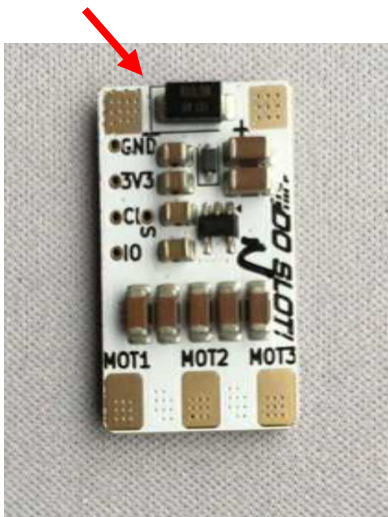
Raivis Edition V1.2 A

Latvian Slot Production V2.10

Latvian Slot Production V2.8

Raivis Edition looks like V2.8 without auto brake bits.

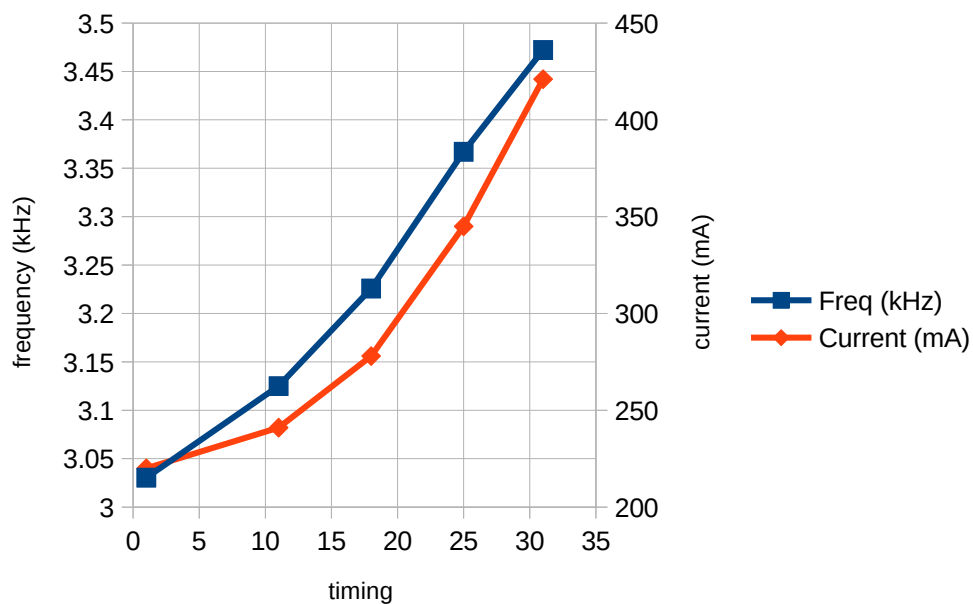
Ruilon SMAJ TVS diode code UV



board	Remora1	Raivis Edition A	Latvian V2.10	Latvian V2.8	Do Slot
weight g	1.45	1.90	2.35	2.31	2.00
pcb dimensions mm, area mm ²	23.21×16.39×0.79 380	23.33×18.74×1.01 437	26.33×20.19×1.04 532	26.29×20.12×1.03 529	24.72×15.95×1.01 394
overall thickness mm	3.09	2.99	2.87	2.95	3.92
smallest pad sizes mm	3.4×1.8	2.2×1.8	2×2	2.5×2	3×3
reverse polarity protection	control & power circuits	control & power circuits	control & power circuits	control & power circuits	control only
auto brake	no	no	yes	yes	no

output mosfets (note thin lead-wire 24AWG is 0.84mΩ/cm)	L0302A (1.2mΩ) Remora1.2	CSD17575 (2.5mΩ) CSD25404 (40mΩ to 5.5mΩ)	CSD16340 (4.2mΩ) CSD25404 (40mΩ to 5.5mΩ)	CSD16340(4.2mΩ) CSD25404 (40mΩ to 5.5mΩ)	CSD16340 (6.1mΩ to 5mΩ) CSD25404 (40mΩ to 5.5mΩ)
reverse polarity mosfet	1R603 (1.2mΩ)	CSD17575 (2.5mΩ)	CSD17575 (2.5mΩ)	CSD17575 (2.5mΩ)	na
start V	1.350	2.348	2.193	1.978	1.957
stop V	1.499	2.02	1.95	1.657	1.955
F at 10.00V 3000Kv motor see graph	T=1, 3.0303kHz T=11, 3.1250kHz T=18, 3.2258kHz T=25, 3.3670kHz T=31, 3.4722kHz	3.4602kHz	3.4247kHz	3.1056kHz	3.4130kHz
no load current mA see graph	T=1, 220 T=11, 241 T=18, 278 T=25, 345 T=31, 421	457	430	284	380
DC bus discharge time with 12R, μs	800	640	810	660	1140
calc capacitance μF	67	53	68	55	95
cap size × nos.	0805 × 10	1207 × 3 (fat 1206)	1206 × 5	1207 × 3 (fat 1206)	1206 × 7

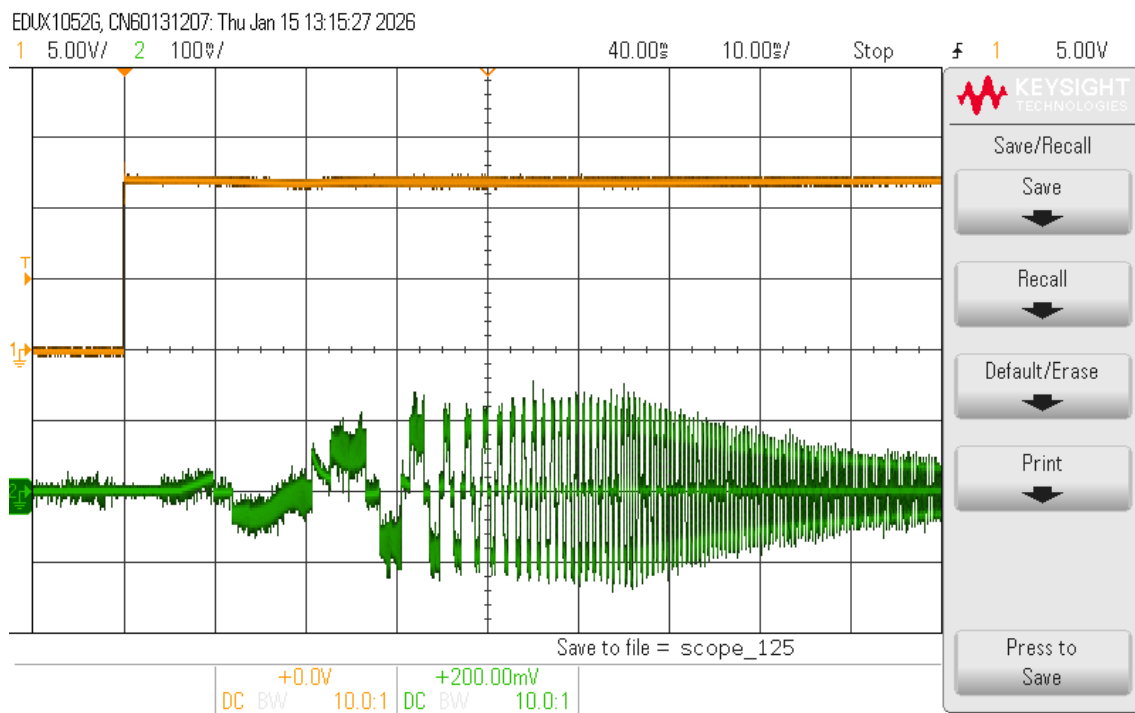
Remora with unloaded 3000Kv motor, rotation frequency & current vs timing



Start table and waveforms

3000Kv motor with flywheel

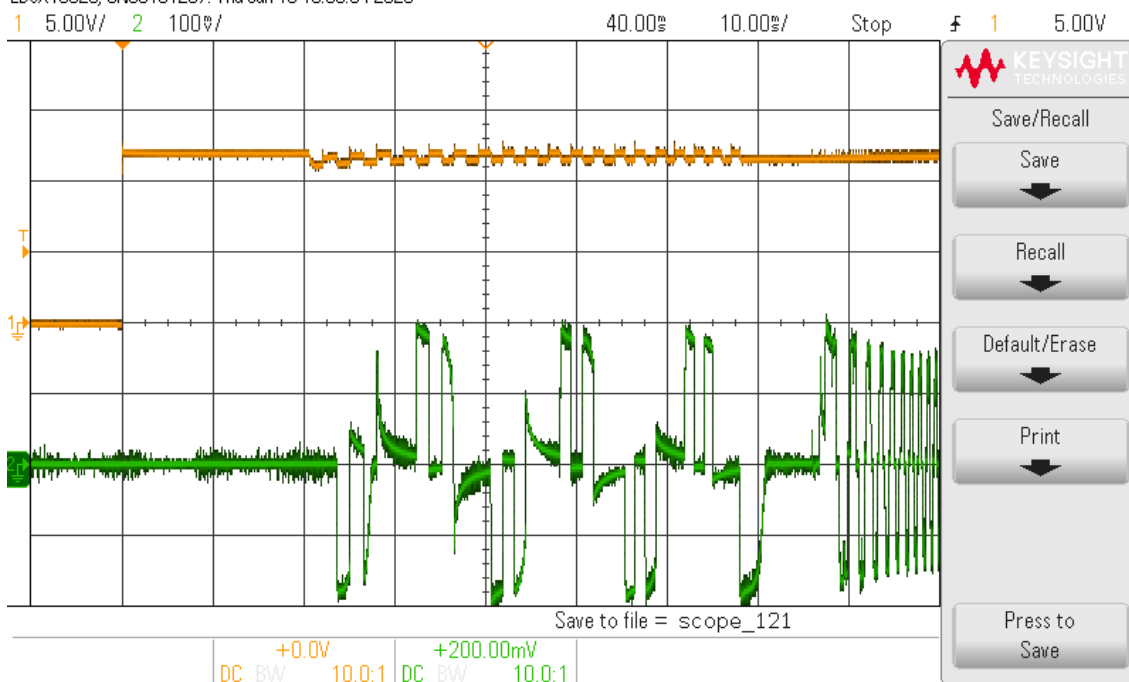
board	Remora1	Raivis Edition A	Latvian V2.10	Latvian V2.8	Do Slot
initial delay ms	6	23	23	23	5
proper rotation at ms	30	75	68	68	would not start
peak current	14	20	20	20	11 about 14A if start at very low voltage initially



Remora with default settings, yellow = supply 5V/div, green = motor current 10A/div, time base 10ms/div.

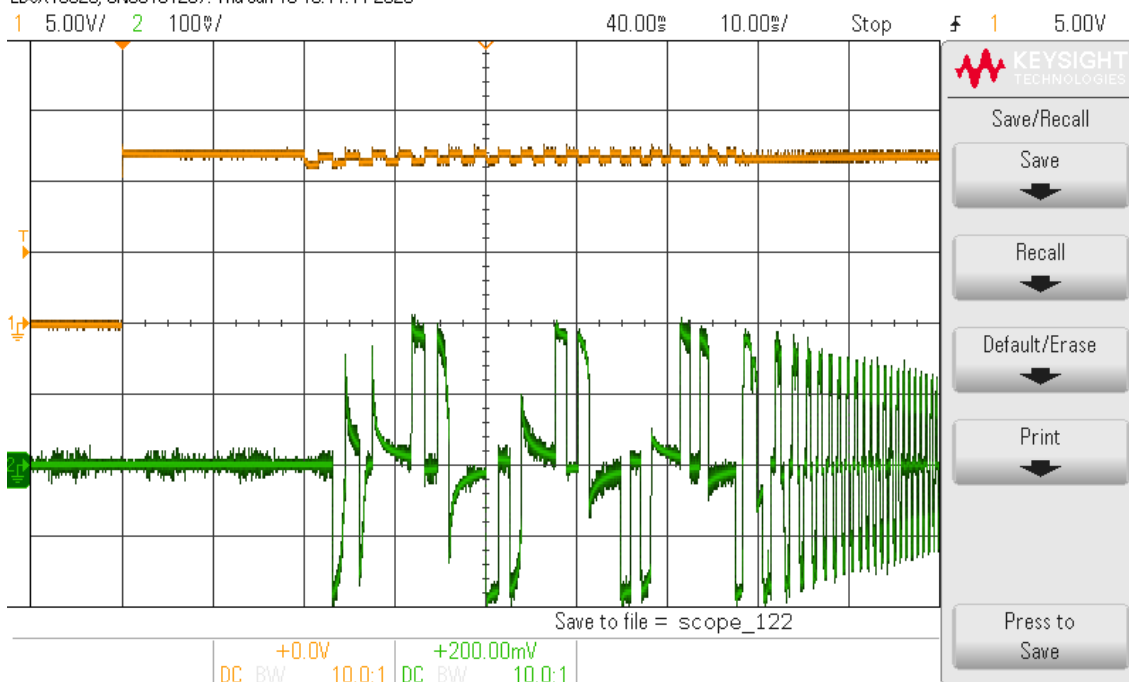
Initial delay ≈ 6 ms, proper rotation ≈ 30 ms, peak current ≈ 14 A,

EDUX1052G, CN60131207: Thu Jan 15 13:09:54 2026



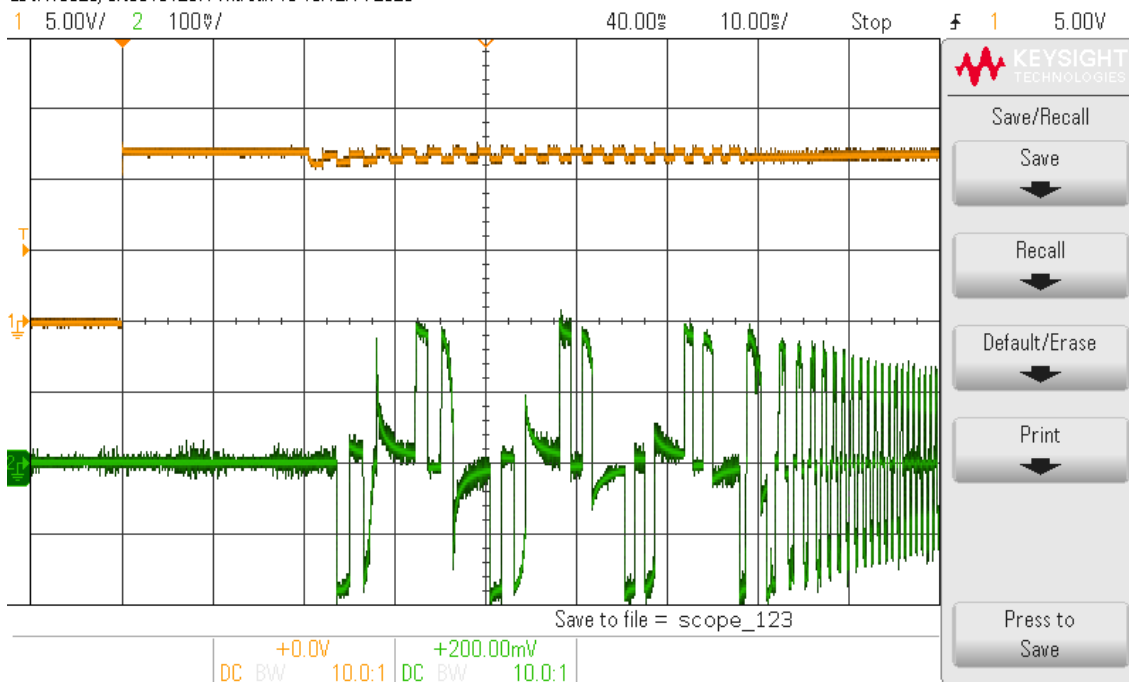
Raivis Edition A, yellow = supply 5V/div, green = motor current 10A/div, time base 10ms/div.
Initial delay $\approx 23\text{ms}$, proper rotation $\approx 75\text{ms}$, peak current $\approx 20\text{A}$,

EDUX1052G, CN60131207: Thu Jan 15 13:11:11 2026



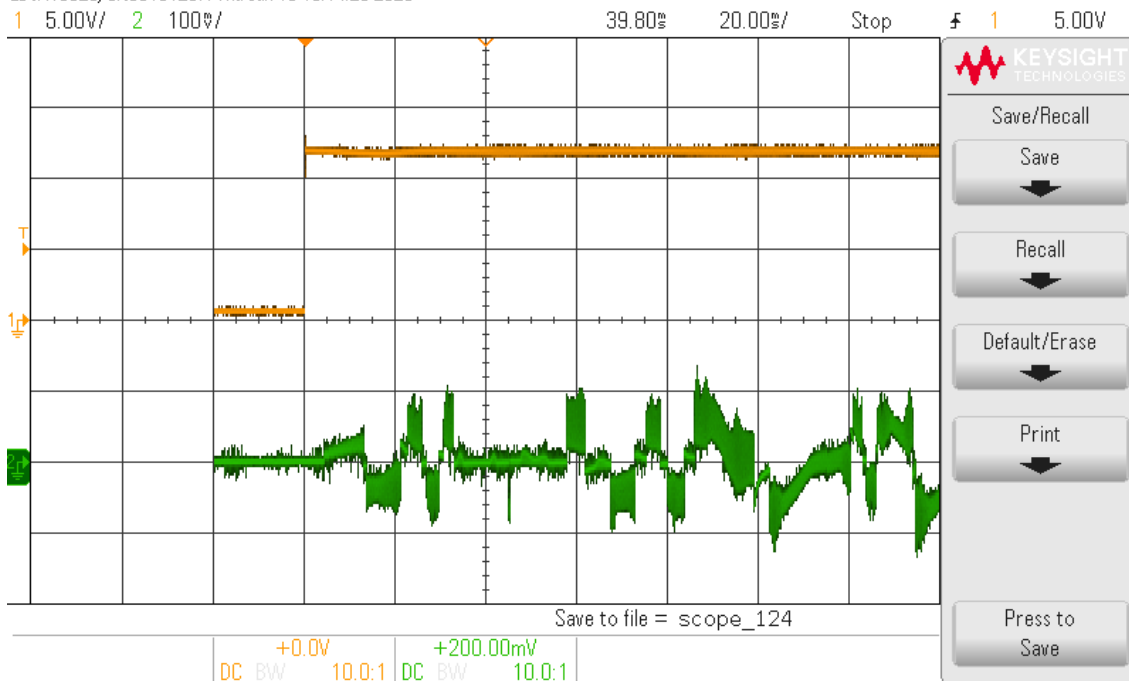
Latvian V2.10, yellow = supply 5V/div, green = motor current 10A/div, time base 10ms/div.
Initial delay $\approx 23\text{ms}$, proper rotation $\approx 68\text{ms}$, peak current $\approx 20\text{A}$,

EDUX1052G, CN60131207: Thu Jan 15 13:12:14 2026



Latvian V2.8, yellow = supply 5V/div, green = motor current 10A/div, time base 10ms/div.
Initial delay $\approx$ 23ms, proper rotation $\approx$ 68ms, peak current $\approx$ 20A,

EDUX1052G, CN60131207: Thu Jan 15 13:14:26 2026



Do Slot, yellow = supply 5V/div, green = motor current 10A/div, time base 20ms/div.
Initial delay $\approx$ 5ms, proper rotation would NOT start, peak current $\approx$ 11A,